JESUS F. SANCHEZ

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EDUCATION

Master of Science - University of Southern California (Los Angeles, CA) Expected May 2026
Aerospace Engineering, Concentration: Propulsion (**GPA: 4.00**)
Bachelor of Science - California State University, Northridge (Northridge, CA) August 2024
Mechanical Engineering, Concentration: Aerospace (**GPA: 3.66**)

WORK EXPERIENCE

NASA Jet Propulsion Laboratory, *Mechanical Engineering Intern* May 2022 - February 2024

- Led the design of complex electromechanical systems from design to manufacturing for Mars' Sample Retrieval Lander (SRL), focusing on design for manufacturability and assembly (DFM & DFA)
- Generated CAD models, engineering drawings using GD&T through Siemens NX and produced written instructions for build, assemble, and test
- Collaborated closely with technicians and the manufacturing team to support fabrication and troubleshoot hardware assembly issues, resulting in accurate and timely delivery of hardware

ENGINEERING PROJECTS & RESEARCH

Liquid Propulsion Laboratory (LPL), *Propulsion Engineer* September 2024 - Present

- Designed a regeneratively cooled 3 kN Jet-A/LOX rocket engine focusing on thrust chamber geometry, nozzle contouring, and injector design, successfully hotfired achieving 95% c* efficiency
- Created a bolt analysis tool to determine the appropriate number of bolts, bolt size, and torque needed to secure the chamber and injector joint sealed under operational loads
- Conducted Finite Element Analysis (FEA) in Ansys Mechanical to assess structural stresses and identify potential sealing issues, resulting in design iterations that reduced gap formation in chamber-injector interface joint
- Performed CFD simulations to guide design iterations of the injector geometry by reducing pressure losses

Evaluation of Novel Scalable Swirl-Flame Hydrogen Combustor, *Research Fellow* May 2023 - Present

- Designed a novel swirl-flame combustor for gas turbine engines capable of operating on hydrogen fuel and presented experimental findings at the Western States Section Combustion Institute (WSSCI) conference
- Performed experimental analysis to study combustor capabilities and process data using MATLAB, resulting in capabilities up to 60% dilution limit
- Conducting CFD analysis through Ansys Fluent to model the flow field and combustion reaction

ASME Human Powered Vehicle (HPV), *Fairing Team Member* August 2023 - May 2024

- Designed, analyzed (CFD), and fabricated a low-drag carbon fiber fairing with a drag coefficient of 0.43, improving from the previous team's 0.5
- Performed finite element analysis (FEA) in SOLIDWORKS Simulation to evaluate the structural integrity of the fairing mounting arms, resulting in a 13% reduction in mass and refined design that reduced stress concentrations
- Presented Conceptual Design Review (CoDR), Preliminary Design Review (PDR) and Critical Design Reviews (CDR) to colleagues and industry professionals

NASA L'SPACE Mission Concept Academy, *Deputy Project Manager* January 2022 - March 2022

- Led a 10-member interdisciplinary team to formulate a lunar rover mission concept for scientific exploration, conducting trade studies and defining mission, vehicle, and subsystem requirements in alignment with NASA's project formulation practices; the concept was evaluated by NASA experts

NASA MINDS Competition, *Mechanical Design Sub-Team Member* September 2020 - May 2022

- 2022 CubeSat: Conducted finite element analysis through SOLIDWORKS to optimize the design of a 4-inch cube satellite structure for weight reduction and material selection
- 2021 Autonomous Robot: Designed the drivetrain system and analyzed (FEA) the chassis of a robot intended for nondestructive testing of space vehicles, enabling it to traverse 90-degree surfaces, achieving 3rd place in competition

SKILLS

Software: **CAD** (Siemens NX, SOLIDWORKS), **CFD** (Ansys Fluent, OpenFOAM), **FEA** (Ansys Mechanical, SOLIDWORKS Simulation), Cantera, LabVIEW, MATLAB, Python, Simulink, Teamcenter
Technical: Geometric Dimensioning and Tolerancing (GD&T), Tolerance Analysis, 3D Printing

COURSEWORK

Fundamentals and Applications of Combustion, Spacecraft Design, Compressible Gas Dynamics, Partial Differential Equations, Rocket Propulsion, Aeropropulsion (see more)